Total No. of Questions : 8]

SEAT No. :

P-597

[Total No. of Pages : 2

[6004]-545

B.E. (E & TC/Electronics)

REMOTE SENSING

(2019 Pattern) (Semester - VIII) (404192C) (Elective - VI)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume any missing data if necessary.***

Q1) a) What is Data Formats and Data Products explain them with their Characteristics? **[8]**

b) Explain Supervised and Unsupervised classification? Compare Supervised and Unsupervised_classification. **[9]**

OR

Q2) a) Explain Basic Principles of Visual Interpretation. Enlist the equipment's used for it. Explain any one in detail. **[9]**

b) Which are Remote Sensing Data Sources? What is USGS and its use? **[8]**

Q3) a) What is Microwave Remote Sensing? Compare Passive and active Microwave Remote Sensing. **[8]**

b) What is Radar Principle? Derive the Radar range equation. **[9]**

OR

Q4) a) What is Side-Looking Airborne Radar Imaging? Explain Ground Range Resolution with figure. **[8]**

b) Explain the Principle of LiDar bathymetry. Explain the types of LiDAR with their applications. **[9]**

Q5) a) Explain GPS Architecture. Also explain Space Segment, Control (Ground) Segment and User Segment in detail. **[9]**

b) What is Satellite based augmentation system? Explain different types of satellite based augmentation system. **[9]**

OR

Q6) a) Illustrate Global Navigation Satellite System. Explain any two GNSS applications in detail. **[9]**

b) Explain in detail the factors that affect GPS performance. Also draw the figure for the same. **[9]**

Q7) a) Enlist applications of Remote Sensing. Explain geological mapping in detail. **[9]**

b) What is watershed and stream network, Explain three steps for water flow management? **[9]**

OR

Q8) a) Draw and explain Flowchart showing Landslide Hazard Modeling using GIS. **[9]**

b) How the Coastal mapping and tsunami simulation is done by using LiDar bathymetry and multi-beam echo sounding. **[9]**

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